

Kepler light curves
67.8 GB, 157,982 FITS files (Q0-Q17)

DR24 TCE catalogue + Autovetter labels
15,683 labelled events, 9,865 stars

NASA Exoplanet Archive
confirmed planets, KOIs, TESS TOIs

1 · Quality-flag masking and gap splitting (segments longer than 0.75 d)

2 · Detrending: cubic B-spline, knot spacing by BIC, transits masked

3 · Phase folding at the reported ephemeris → nine views and derived scalars

4 · Star-level 80 / 10 / 10 split — the same rows for every model

5 · Training — PHANTOM, AstroNet, GBDT, classical cascade; five seeds each (61 runs)

6 · Evaluation — average precision, AUC, precision at 95% recall; Welch t-test

Ablations
five seeds per configuration

Conformal + Benjamini-Hochberg
candidate lists with an FDR bound

Transit-parameter recovery
zero-shot transfer to TESS